

DANIEL ADEJUMO

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EDUCATION

New York University

New York, USA

M.S. in Mechatronics and Robotics Engineering (CGPA: 3.63/4.0)

September 2024 - May 2026

Relevant Coursework: Reinforcement Learning & Optimal Control, Mathematics for Robotics, Advanced Mechatronics

University of Ibadan

Ibadan, Nigeria

B.Sc. Mechanical Engineering (First Class Honors, CGPA 3.70/4.0)

June 2023

B.Sc. Thesis: UI Hand I - The Modelling, Design, and Control of a Dexterous Robotic Hand

RESEARCH INTEREST

Dynamics and Control, First-Order Model-Based Reinforcement Learning, and constrained RL for robust and safe control of high-dimensional robotic systems.

PUBLICATIONS

K. Runwal, S. Gajare, **D. Adejumo**, O. Ankalkope, S. Baroth, and A. Arab, "A Semantic Observer Layer for Autonomous Vehicles: Pre-Deployment Feasibility Study of VLMs for Low-Latency Anomaly Detection," arXiv:2603.28888, 2026.

Adejumo, D.O., Fadare, D.A., Kazeem, R.A., Ikumapayi, O.M., Falana, A., Adedayo, A.S., Fadare, D.A., Adeoye, A.O.M., Ogundipe, A.T., Olarinde, E.S. (2023). A low-cost, modular, cable-driven, anthropomorphic robotic hand: A conceptual design and application in biomimetic study. Journal Européen des Systèmes Automatisés, Vol. 56, No. 4, pp. 627-639. <https://doi.org/10.18280/jesa.560412>

SKILLS

- **Programming and Software:** Python, C/C++, MATLAB, CUDA, JavaScript, OpenCV, Git, Linux, Docker
- **Machine Learning:** Deep Learning, Deep Reinforcement Learning, LLMs, VLMs, PyTorch, HuggingFace
- **Robotics and Embedded Systems:** Electronics, Microcontroller Programming, Sensor/Actuator Interfacing, PID and MPC Control, Robot Operating System (ROS), Gazebo, Simscape, MuJoCo, IsaacSim & Isaac Lab
- **Design (CAD) and Rapid Prototyping:** Fusion 360, SolidWorks, AutoCAD, 3D Printing, CNC Machining
- **Soft Skills:** Communication, Teamwork, Project Management, Problem-solving

PROJECTS

Anomaly Detection for Autonomous Vehicles

September 2025 – December 2025

- Built a video-based anomaly detection pipeline for Autonomous Vehicles (AVs) using Nvidia Cosmos-Reason1 (Qwen2.5-VL-7B-Instruct) Vision-Language Model (VLM).
- Optimized the inference pipeline and applied prompt tuning and pruning, reducing reasoning time by 80%.
- Implemented FP4 quantization for deployment on an RTX 5090, achieving ~500 ms end-to-end reasoning for real-time video anomaly detection while maintaining accuracy.

Reward Shaping for Quadruped Locomotion

November 2025 – December 2025

- Trained a command-conditioned quadrupedal locomotion policy using PPO for the Unitree Go2 robot in IsaacLab, leveraging NYU's high-performance computing (HPC) infrastructure.
- Designed and optimized reward functions to produce stable, natural-looking, and robust gaits; extended the same framework to enable backflip with recovery and bipedal (hind-leg) locomotion on the Go2.
- Applied domain randomization across contact friction, inertial properties, and actuator dynamics to improve robustness and reduce the sim-to-real gap.

Reinforcement Learning for Gait Control in Bipedal Robots

January 2025 – April 2025

- Trained velocity-conditioned locomotion controllers for Unitree H1 and G1 bipedal robots in the Genesis physics engine using reinforcement learning (PPO), enabling dynamic motion control.
- Applied advanced reward engineering techniques to improve gait stability and policy robustness.
- Initiated Sim2Real policy transfer on a Unitree G1 to evaluate policy generalization to the real world.

Design and Control of an Anthropomorphic Robotic Hand (B.Sc. Thesis)

February 2022 – June 2023

- Designed and prototyped a 26-DOF biomimetic robotic hand with tendon-driven actuation, and applied Soft Actor-Critic (SAC) and Proximal Policy Optimization (PPO) for fingertip pose control.
- Engineered the embedded control architecture, including actuator integration, microcontroller communication protocols, and low-level motor control using C++ and Python on Raspberry Pi and Pico MCUs.
- Prepared and reviewed an academic paper, including a comprehensive literature review, collaborated with

co-authors, and published the paper in the Journal Européen des Systèmes Automatisés.

Autonomous Mobile Drawing Robot

January 2025 – April 2025

- Designed a mobile robotic system with an integrated pen mechanism to overcome workspace limitations of fixed-base automated drawing platforms, utilizing Arduino and Propeller microcontrollers for control.
- Developed an image-to-G-code processing pipeline in C++ and Python, using computer vision techniques (OpenCV) to convert raster images, SVG, and DXF formats into the mobile robot's motion commands.
- Designed detailed CAD models in Fusion 360 and fabricated the system with 3D printing.

Course Projects (Foundation of Robotics and Mechatronics)

September 2024 – December 2024

- Implemented forward and inverse kinematics for a 6-DOF robotic arm in MATLAB, solving nonlinear equations and validating solutions through MATLAB-based visualizations.
- Designed and integrated a mobile robotic system with an articulated arm for autonomous detection and removal of liquid spills and trip hazards in household environments.
- Repurposed a capacitive moisture sensor as a spill detector by leveraging capillary action in foam materials.

EXPERIENCE

Tuesday Labs

New York, USA

Locomotion Intern

January 2026 – May 2026

- Implemented a temperature-aware RL policy by integrating actuator thermal models into simulation and reward terms, enabling safe actuator operating temperatures during real-world robot deployment.
- Extended, trained, and deployed an RL motion tracking pipeline for bipedal robots, adding custom and bi-directional inference playback to increase the tasks performable from a single motion reference and policy.
- Developed a variable base-height bipedal locomotion controller using Centroidal Dynamics, achieving 1.5 m/s walking speeds and robustness to 50 N perturbations in MuJoCo and Newton.

New York University

New York, USA

Graduate Teaching Assistant (Advanced Mechatronics)

January 2026 – May 2026

- Provided technical support to 6 student teams during project ideation and implementation for mid-semester and term projects.
- Evaluated implementations of weekly take-home, hands-on assignments, providing technical feedback to accelerate students' understanding and practical skills.
- Advised students on course content and questions during office hours.

Course Assistant (Innovation & Science of Smart Cities)

July 2025 – August 2025

- Prepared a detailed 5-week program and class activities for the Innovation, Entrepreneurship and Science of Smart Cities (IESoSC) summer program at the NYU Center for K12 STEM Education.
- Delivered class instructions to 25 students, supported students with hands-on class activities, and supervised students' projects to prepare them for the Shark Tank-style showcase at the end of the program.
- Collaborated with other class instructors, managed students' data, and communicated class progress and requirements to program managers to ensure a seamless and successful program.

Soft Quest Incorporated (SQI) College of ICT

Ibadan, Nigeria

Automation and Robotic Systems Engineer

September 2022 – August 2024

- Trained students to acquire foundational skills in robotics, including electronics design, embedded systems, programming, Computer-Aided Design (CAD), and rapid prototyping (3D printing, CNC machining).
- Collaborated with other teams and departments at SQI to organize events, including a 1-week robotics training bootcamp welcoming over 100 students from nearby institutions.
- Installed and maintained equipment, including 3D Printers, CNC Laser Cutting Machines, and PCB Printers.

Control Systems & Simulation Intern, Robotics Department

August 2021 – February 2022

- Developed and validated the powertrain subsystem for the Alpha Electric Vehicle using model-based design in MATLAB and Simulink, applying control system principles to optimize dynamic performance and accelerate design iteration cycles.
- Delivered design insights to technical and non-technical stakeholders through structured presentations and simulation-driven visuals, supporting clear decision-making throughout the project lifecycle.
- Designed and prototyped embedded automation systems, including a gas metering system and security alert system, using sensor integration and microcontroller logic.

Nigerian Institution of Mechanical Engineers, University of Ibadan

Ibadan, Nigeria

Team Lead, Control Electronics Subsystem – Project Helios

December 2019 – May 2020

- Led a 6-member team to develop the electronics and control subsystems for a student electric vehicle project, overseeing system design, component selection, and integration strategy.
- Defined design requirements for the powertrain and electrical systems, ensuring alignment with control

objectives and overall vehicle architecture.

AWARDS

- Students' Choice Award, NYU Tandon Capstone Project Competition (Team Award, April 2025)
- SEED Scholarship, New York University Tandon School of Engineering (2024 – 2026)
- Dean's Honor Awards, Faculty of Technology, University of Ibadan (5x, 2017 – 2023)
- Outstanding Cambridge Learner Award, Top in Nigeria, Physics (November 2016)